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# Atomic Force Microscope Studies of Membranes: Surface Pore Structures of Diaflo Ultrafiltration Membranes

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**Noncontact atomic force microscopy (AFM) has been used to investigate the surface pore structure of eight Diaflo ultrafiltration membranes covering a range of nominal molecular weight cutoff (MWCO) from 3000 to 300,000 and manufactured from three different polymer types. Excellent high resolution images were obtained. Analysis of the pore images gave quantitative information on the surface pore structure, in particular the pore size distribution and surface roughness. Such data is compared to that obtained by other techniques. Noncontact AFM is a facile and informative means of studying the surface structure of porous materials such as synthetic membranes.** © 1996 Academic Press, Inc.

*Key Words:* **atomic force microscopy; ultrafiltration; pore size distribution.**

## 1. INTRODUCTION

The use of membranes to separate components of a process flow is arguably one of the most important recent developments in chemical engineering. Membrane processes are advanced filtration processes which utilize the separation properties of finely porous polymeric or inorganic films. Of the wide range of membrane processes (1), the pressure driven process known as ultrafiltration is one of the most widely used. Ultrafiltration is usually defined as being a process separating materials in the size range 0.1 $\mu$m–5 nm. The effectiveness of ultrafiltration as a separation process will clearly depend on the pore size distribution of the membrane being used. Surface roughness is also a significant property as it has a strong influence on the deposition of process stream components on the top surface of the membrane. Hence, in developing membrane processes it is important to have knowledge of surface pore structures.

The techniques which have been most widely used for the study of membrane morphology and pore characteristics are electron microscopy, mercury intrusion, extended bubble point, gas adsorption–desorption, thermoporometry, solvent permeability, and solute challenge (2). Of these, only electron microscopy can provide direct and detailed information on the size distribution, shape, and topography of the pores. However, electron microscopy has a number of disadvantages in the examination of porous polymeric structures. These include the fact that it is a high vacuum technique (which results in extensive drying of the sample), the requirement for extensive preparation such as metal coating or formation of a replica (both of which can produce artifacts with finely porous materials), and the possibilty of damage by the electron beam. It is for reasons such as this that the use of atomic force microscopy (AFM) is so tantalizing to membrane technologists. Atomic force microscopy (3) gives topographic images by scanning a sharp tip over a surface (4) and has been used to produce atomic resolution images of both conductors and nonconductors (5) including biological materials (6). AFM can image nonconducting surfaces with nanometer scale resolution in air and even under liquids. Consequently, the samples need not be exposed to vacuum and preparation techniques like evaporating a thin metal coating or taking a replica are unnecessary. Even soft organic surfaces can be successfully imaged with AFM because the use of microfabricated cantilevers (5) allows operation with total forces between $10^{-7}$ and $10^{-8}$ N in contact mode and on the order of $10^{-12}$ N in noncontact mode.

Atomic force microscopy has been used to investigate microfiltration and ultrafiltration membrane surface structure (7–15). AFM has also been used to elucidate the mechanisms giving rise to inefficiencies (fouling) in membrane processing (16). However, with one exception (14), membranes have been studied in ''contact mode.'' In this mode, also known as repulsive mode, the tip is very close (in contact) to the surface being imaged and is responding to very short range repulsive interactions with the sample. In a second and more recently developed mode of operation, ''noncontact mode,'' the tip instead responds to attractive van der Waals interactions with the sample. These are longer range, so the tip is generally at a distance of 5–10 nm away from the surface while imaging. This mode of operation is especially suitable for materials which are soft or liable to mechanical damage as the forces used for imaging are also lower. Polymeric membranes are one such class of material. Further, it has been reported (14) that use of noncontact mode can in some cases allow imaging of membrane surfaces which cannot be imaged in contact mode. However,

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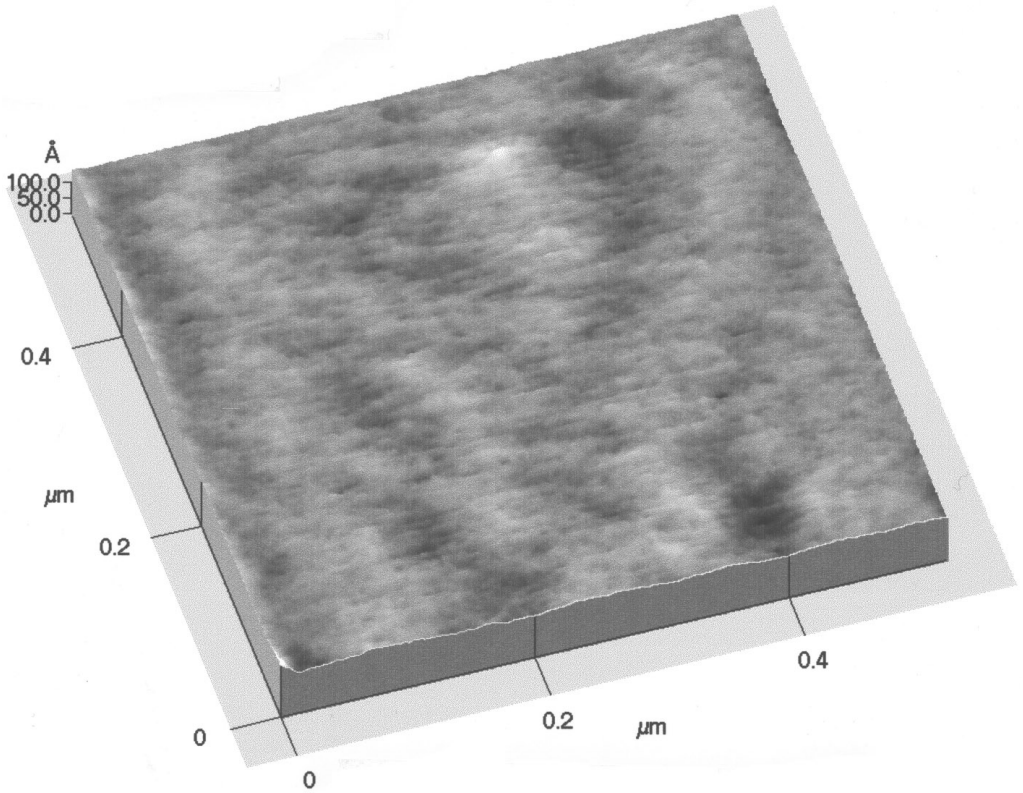



**FIG. 1.** AFM image of YM3 Diaflo ultrafiltration membrane (Reproduced by permission of University of Wales Swansea).

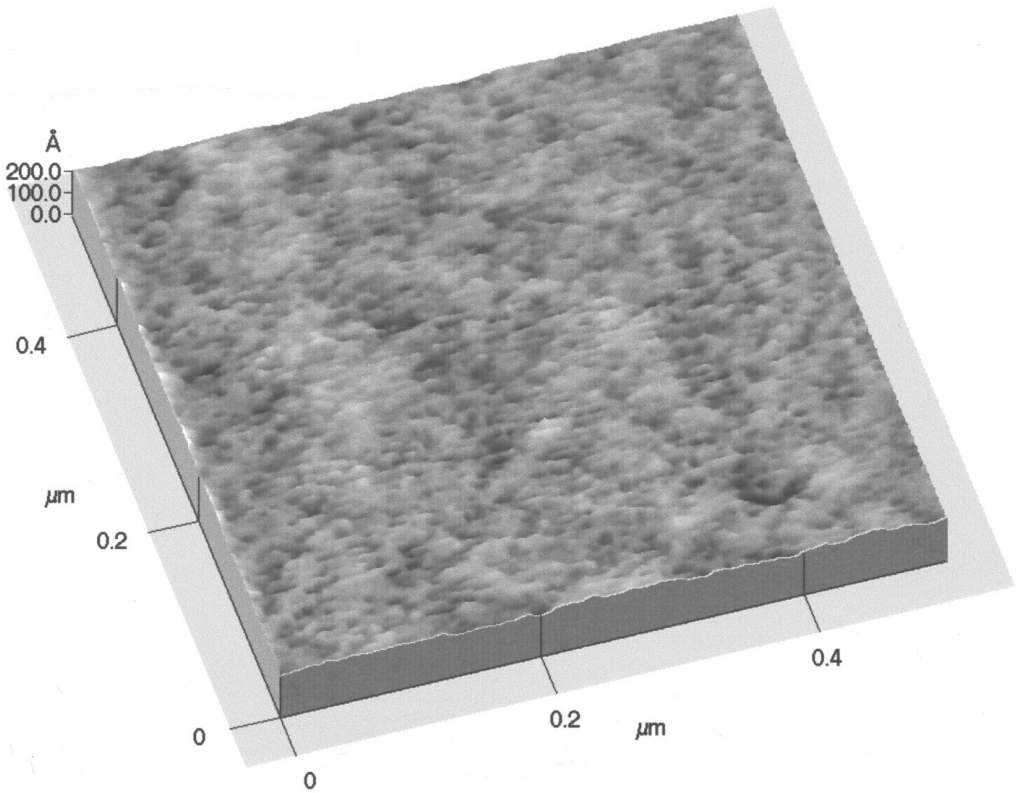



**FIG. 2.** AFM image of YM10 Diaflo ultrafiltration membrane (Reproduced by permission of University of Wales Swansea).

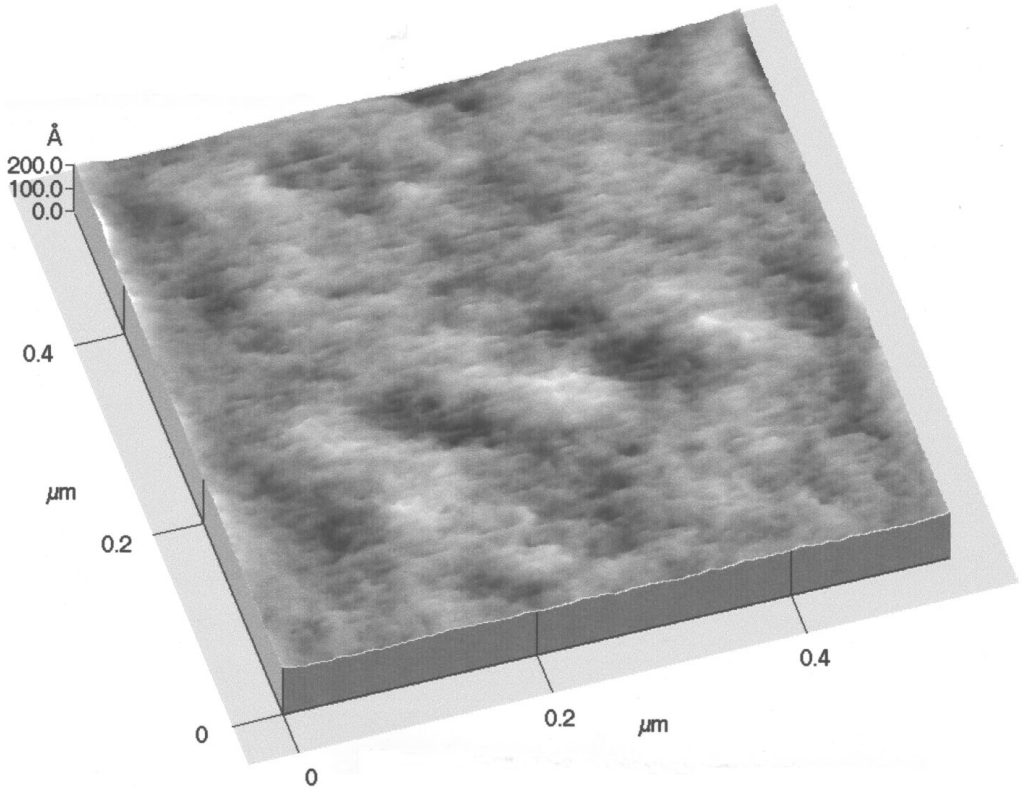



**FIG. 3.** AFM image of YM30 Diaflo ultrafiltration membrane (Reproduced by permission of University of Wales Swansea).

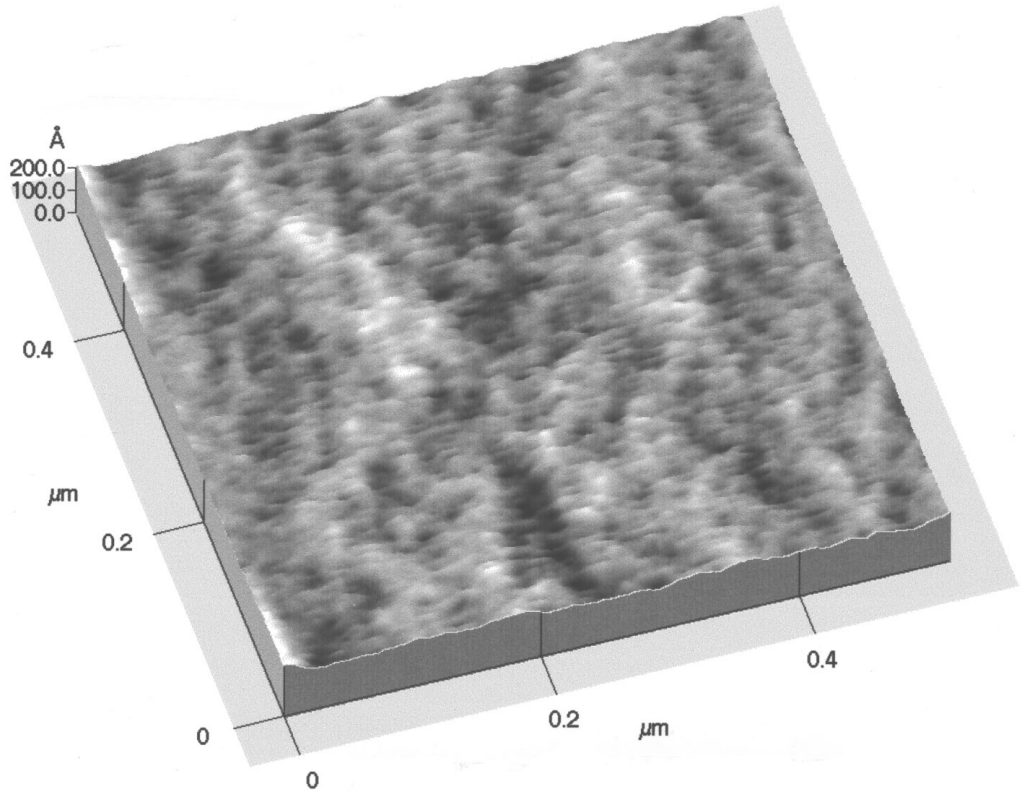



**FIG. 4.** AFM image of YM100 Diaflo ultrafiltration membrane (Reproduced by permission of University of Wales Swansea).

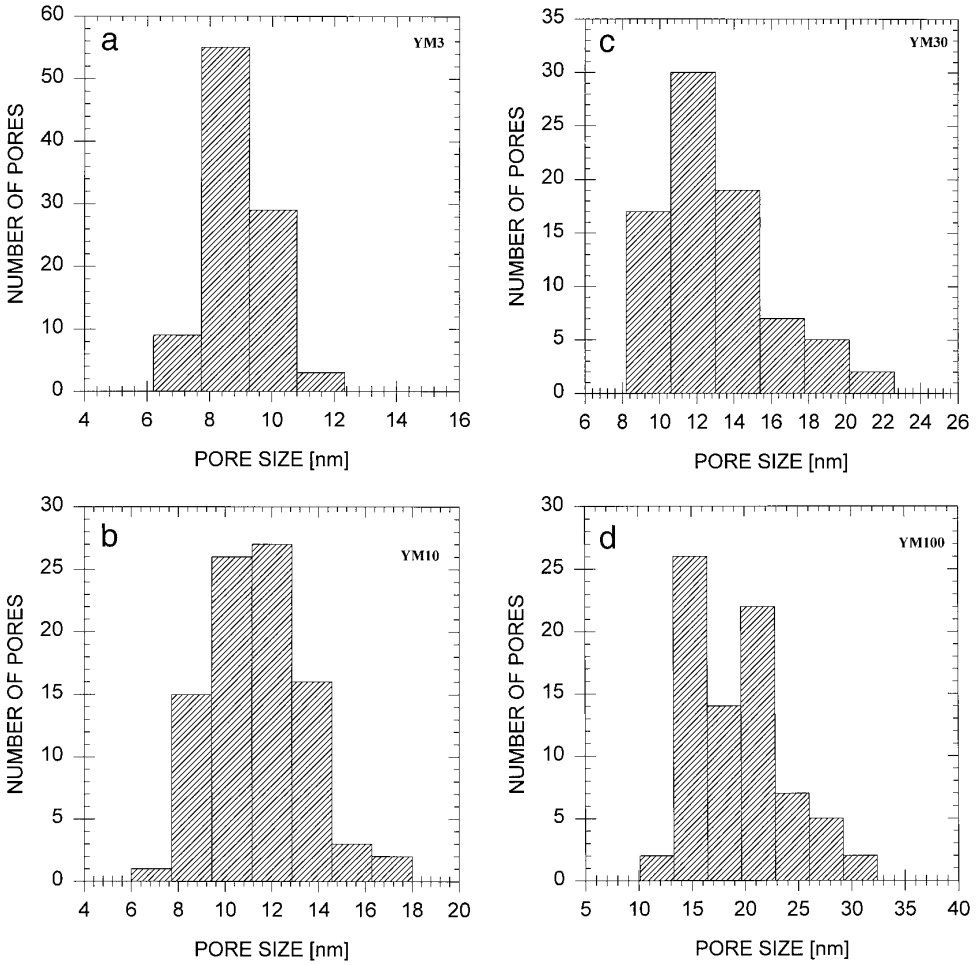



**FIG. 5.** Pore size distributions of YM Diaflo ultrafiltration membranes (a) YM3, (b) YM10, (c) YM30, and (d) YM100.

the systematic use of noncontact AFM for quantifying the surface pore structure of membranes has not been reported previously.

The present paper reports a study of the use of noncontact AFM to investigate the surface pore structure of Diaflo ultrafiltration membranes covering a wide range of molecular weight cutoff (MWCO), and hence likely pore size, and manufactured from three different polymer materials. Quantitative measurements of pore size, pore size distribution, and surface roughness have been obtained. The results show that noncontact AFM is an excellent means of obtaining such information for ultrafiltration membranes.

## 2. MATERIALS AND METHODS

The AFM used in the present study was an Autoprobe (CP-100), a commercial device from Park Scientific Instruments (USA). High magnification images of membrane surfaces can be obtained with the use of microfabricated cantilevers (5). In the present work, Ultralevers (Park Scientific Instruments) were used. These are silicon cantilevers with a high aspect ratio tip of typical radius of curvature 10 nm. Noncontact AFM measures long range attractive van der Waals forces between the tip and the sample—the tip is held about 5 to 10 nm above the sample surface during the scan. As these forces are much weaker than the repulsive forces measured in contact mode, a highly sensitive modulation technique is used to detect the small force gradients. A stiff cantilever is vibrated near its resonance frequency (typically 100–400 kHz) with an amplitude of a few nanometers, and changes in the cantilever resonance are detected as the tip is scanned above the sample surface. While scanning the sharp tip above the surface, the vertical motion of the tip due to interatomic forces, is detected by sensing, with a two segment photodiode, the deflection of a laser beam reflected off the back of the cantilever. In operation, the sample is moved under the tip while a z-feedback loop operates to maintain a constant force on the cantilever tip (a constant vibration amplitude) by adjusting the tip-to-sample spacing. The z voltage applied to the (piezoelectric) scanner to maintain constant attractive force is used to map out the surface topography. A scan rate of 1 Hz was used with 256 $\times$ 256 pixel resolution. No distortion of the images

**TABLE 1**
**Statistical Characterization of Diaflo Membranes**

| Diaflo membrane | YM3 | YM10 | YM30 | YM100 | PM10 | PM30 | XM50 | XM300 |
|---|---|---|---|---|---|---|---|---|
| Molecular weight [MW] | 3000 | 10,000 | 30,000 | 100,000 | 10,000 | 30,000 | 50,000 | 300,000 |
| Mean size [nm] | 8.7 | 11.3 | 13.2 | 19.4 | 16.4 | 30.6 | 87 | 147.2 |
| Standard deviation [nm] | 1.3 | 2.4 | 2.8 | 4.2 | 3.9 | 9.6 | 20.4 | 42.7 |
| Size range [nm] | 6.2–12.3 | 6.1–18.0 | 8.2–22.6 | 10.1–32.4 | 7.7–25.2 | 16.7–62.7 | 46.4–147.1 | 77.6–279.5 |
| Number of counted pores | 96 | 90 | 80 | 78 | 57 | 58 | 99 | 77 |
| Rms. rough [nm] | 0.36 | 0.53 | 0.6 | 0.77 | 0.87 | 0.8 | 2.8 | 13.4 |

due to condensation between tip and sample was noted in the present work.

In the present work, the AFM was used to image eight Diaflo (Amicon) ultrafiltration membranes. These membranes were chosen due to their wide availability and application and since they cover a wide range of MWCO (17) and are manufactured from three different polymers. The membranes used were:

—YM3, YM10, YM30, and YM100, with MWCO of 3000, 10,000, 30,000, and 100,000, respectively, made from regenerated cellulose,

—PM10 and PM30, with MWCO of 10,000 and 30,000, respectively, made from polysulfone,

—XM50 and XM300, with MWCO of 50,000 and 300,000, respectively, made from Dynel (poly *co*(acrylonitrile-vinyl chloride).

The manufacturer defines the molecular weight cutoff as the molecular weight of a globular solute at which the solute is 90% rejected by the membrane. Further membrane specification is given in the Amicon catalog.

The only preparation of the membranes required was their attachment to steel disks with double-sided Scotch tape. The measurements were made directly in air operating in noncontact mode.

For the analysis of surface pore characteristics the AFM image processing program was used. This allows work with different color tables and adjustment of the contrast and image representation in top view and in perspective. In addition, its ability to generate line profiles along selected lines in the images was utilized for the analysis of single pores and region profiles.

## 3. RESULTS AND DISCUSSION

### *3.1. YM Series*

Figures 1–4 show AFM images of the YM3, YM10, YM30, and YM100 membranes. The images are shown in three-dimensional form over areas of $0.5 \times 0.5$ $\mu$m. The color intensity shows the vertical profile of the samples, with the light regions being the highest points and the dark regions the depressions and pores. The pores are clearly visible as small well-defined dark areas on the images. In some cases they appear to occur in clusters, and for the higher MWCO membranes there is an increasing tendency for the pores to occur at the bottom of depressions in the membrane surface. The increase in pore size with increasing MWCO of the membranes is immediately apparent on comparing the images. There is also a visually apparent increase in surface roughness as the MWCO increases.

The AFM software allows quantitative determination of the diameter of pores by the use of the images in conjunction with digitally stored line profiles (16). Pore size distributions obtained by analyzing large numbers of pores in this way are shown in Fig. 5. Table 1 presents statistical information on the mean size, standard deviation, and size range. It may be seen that the mean pore size increases systematically as the specified MWCO of the membranes increases with a relatively small standard deviation in all cases. The root mean square roughness also increases as the nominal molecular weight cutoff increases, though the magnitude is in all cases relatively low.

### *3.2. PM Series*

Figures 6 and 7 show images of the PM10 and PM30 membranes. The images are shown in three-dimensional form over areas of $1.0 \times 1.0$ $\mu$m. Again the pores are clearly visible as dark areas on the images. For these membranes there is a greater tendency for the pores to occur in clusters at the bottom of depressions. This may be seen, for example, by comparing Fig. 2 (YM10, MWCO 10,000) with Fig. 6 (PM10, MWCO 10,000). The pores in the YM membrane have the more uniform distribution and with a much increased tendency to occur discretely rather than in clusters.

Pore size distributions for the PM membranes are given in Fig. 8 and statistical analysis of the pores is given in Table 1. Both the PM10 and PM30 membranes have a significantly larger mean pore size than the YM membranes of the same specified MWCO. In this context, it should be remembered that rejection at membranes depends not only on size but also on specific interactions of a colloidal nature, such as electrical charge and dispersion forces (18). The root mean square roughness is also greater for the PM membranes than for the YM membranes of comparable MWCO.

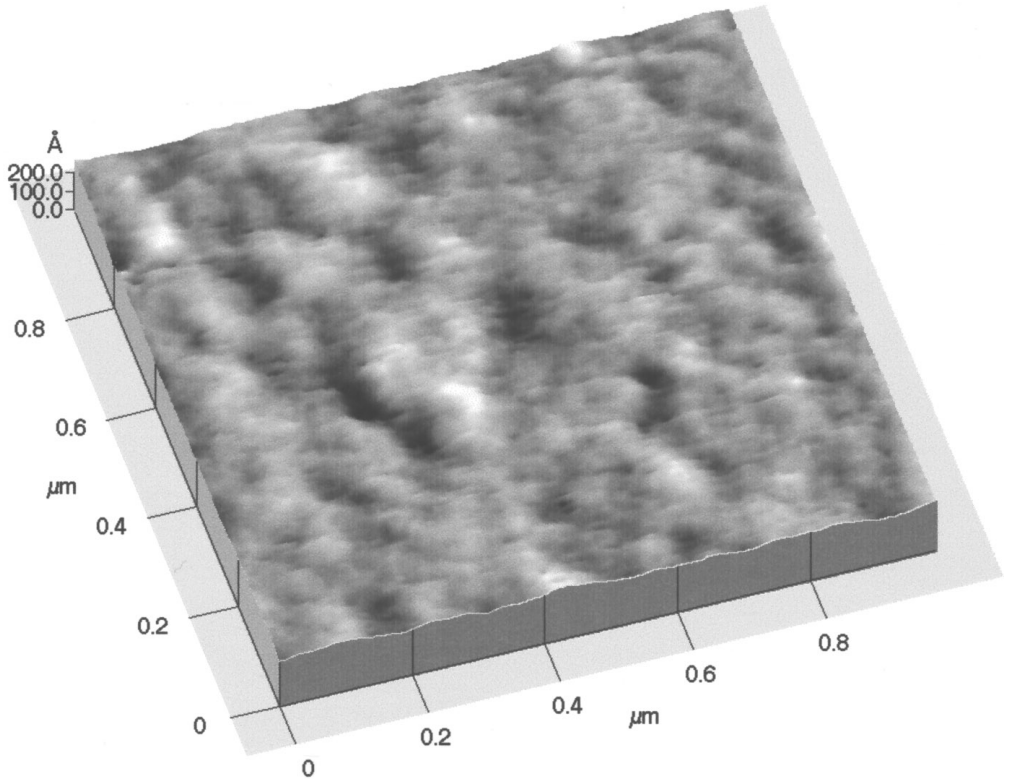



**FIG. 6.** AFM image of PM10 Diaflo ultrafiltration membrane (Reproduced by permission of University of Wales Swansea).

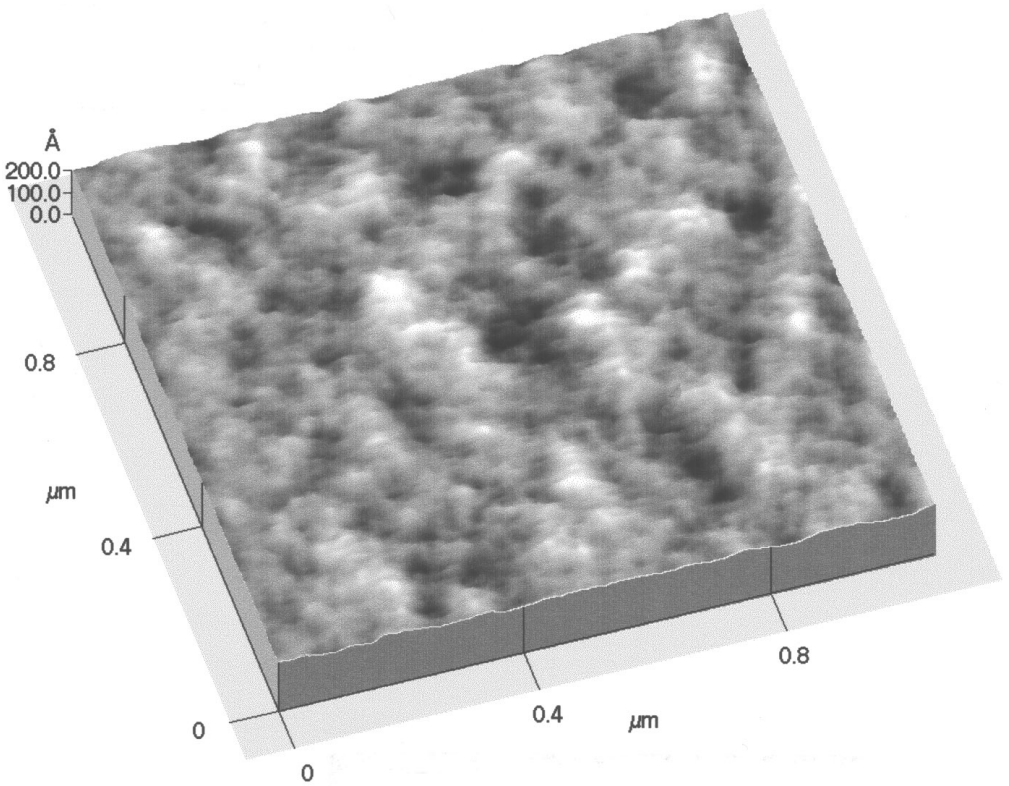



**FIG. 7.** AFM image of PM30 Diaflo ultrafiltration membrane (Reproduced by permission of University of Wales Swansea).

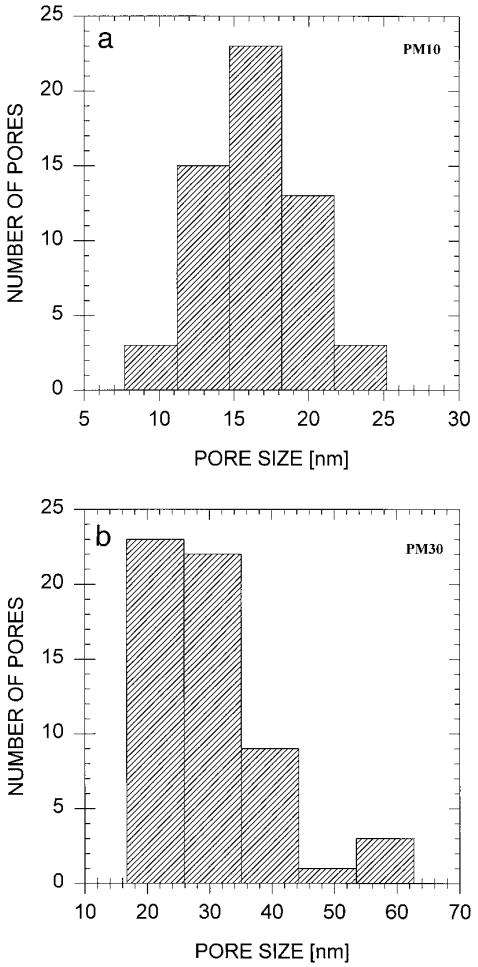



**FIG. 8.** Pore size distributions of PM Diaflo ultrafiltration membranes (a) PM10 and (b) PM30.

### *3.3. XM Series*

Figures 9 and 10 show images of the XM50 and XM300 membranes. The images are shown in three-dimensional form over areas of 2.0 × 2.0 $\mu$m. The range of specified MWCO is greater for these membranes than for the other series, and the comparatively large size of the pores is clear. This is confirmed by the pore size distributions given in Fig. 11 and the statistical information of Table 1. Indeed, the mean pore size of 0.147 $\mu$m determined for XM300 would lead to this membrane being classified as a microfiltration membrane rather than an ultrafiltration membrane. The table shows that the root mean square roughness of these membranes is substantially greater than for either the YM or PM series.

### *3.4. Comparison with Other Data*

In previous investigations of membranes using AFM, there have been a few systematic measurements of pore size distribution, though none using noncontact AFM. So the first point of comparison of the present data is with pore size distributions determined by contact AFM. The most systematic previous investigation (10) included data for three membranes with specified MWCO 10,000, one with MWCO 40,000 (polyethersulfone) and three of MWCO 100,000 (one polysulfone and two polyethersulfone). These membranes were from different manufacturers to those of the present work. For the 10,000 MWCO membranes, mean pore sizes of 14.1, 12.6, and 18.8 nm were reported. Comparison with the data for YM10 and PM10 in Table 1 shows that such results are consistent with the present data. For the 40,000 MWCO membrane, a mean pore size of 11.6 nm was reported which is comparable to the present results for YM30 but smaller than the present results for PM30. For the 100,000 MWCO membranes, mean pore sizes of 22.1, 25.2, and 26.2 nm were reported, which is comparable to but greater than the value obtained for YM100 in the present work. The width of pore size ranges reported previously are comparable to those in Table 1.

Other published pore size data from AFM has been sparse, though these include the ranges 15–25 nm and 50–60 nm for polyethersulfone membranes of MWCO 30,000 and 100,000, respectively (11), 7–9 nm for a 10,000 MWCO polyethersulfone membrane (12), and 12–20 nm for a 30,000 MWCO polyacrylonitrile membrane (14). Comparison with the data of Table 1 again shows that previous AFM studies are broadly consistent with the present work.

The second important type of comparison which can be carried out is with the results from electron microscopy. As mentioned in the Introduction, electron microscopy data for membranes can require careful interpretation due to the extensive sample preparation required. One of the most detailed and careful studies has used a high resolution field emission scanning electron microscope (FESEM) (19), which uses a very low accelerating voltage (2–5 kV) thus minimizing damage to the membranes. Some of the YM30, YM100, PM30, and XM300 membranes used in the present work were also included in the present study. A comparison was also made with data from transmission electron microscopy (TEM).

Even operating the FESEM with the lowest energy electron beam (2 kV), it was found that the YM membranes were damaged, so no quantitative data could be obtained for these membranes (19). Hence, no direct comparison with the data of Table 1 is possible, though the fact that the YM membranes could be imaged by noncontact AFM shows the utility of the technique. For the PM30 membrane, FESEM gave a mean pore size of 4.0 nm and a range of 1.5–9.8 nm, while for the XM300 membrane a mean pore size of 8.6 nm was found with a range of 2.1–22.3 nm (19). These values are significantly smaller than those obtained by AFM. Reference (19) also tabulates TEM data from those authors and other workers. There is wide variation in such data. Hence, the TEM data tabulated for PM30 gives mean pore sizes of 5.5 and 11.8 nm with respective ranges 1.1–10.1 nm and 6–29 nm. For XM300 means of 7.7, 20.0, or 24.5

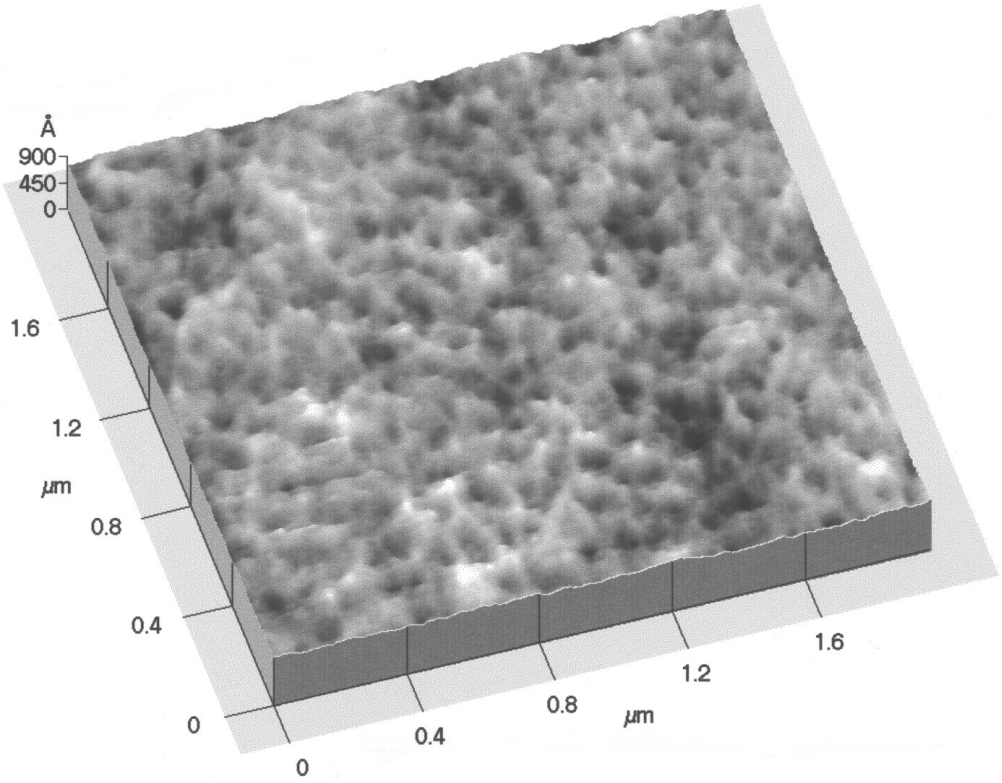



**FIG. 9.** AFM image of XM50 Diaflo ultrafiltration membrane (Reproduced by permission of University of Wales Swansea).

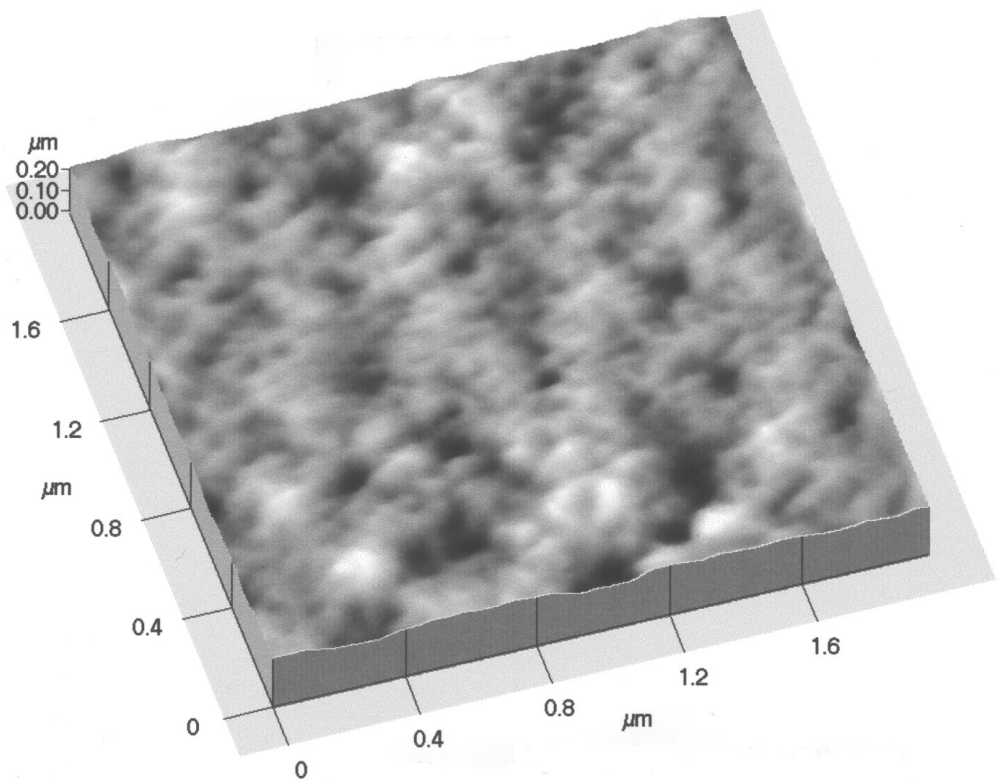



**FIG. 10.** AFM image of XM300 Diaflo ultrafiltration membrane (Reproduced by permission of University of Wales Swansea).

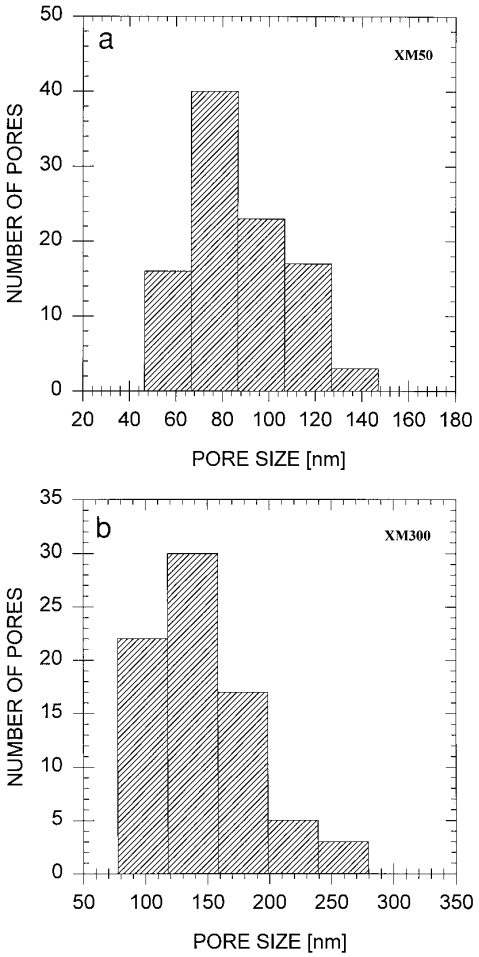



**FIG. 11.** Pore size distributions of XM Diaflo ultrafiltration membranes (a) XM50 and (b) XM300.

nm are reported with respective ranges 1.9–17.5 nm, 8–47 nm, and 13.0–37.5 nm. Despite the spread of these means, they are still smaller than those determined by noncontact AFM in the present work, though the AFM data agrees better with TEM data than with FESEM data.

The noncontact AFM data in the present work is internally consistent and is in good agreement with previous contact AFM studies of membranes. The discrepancies between pore sizes determined by AFM and EM are not easily resolved. SEM requires deposition of a conducting coating on the sample and TEM requires the preparation of a replica. The lengthy preparation techniques used in both cases can produce artifacts. Both coatings and replica formation are likely to lead to underestimation of pore sizes. This is confirmed by the finding that hydraulic permeabilities calculated from EM pore dimensions are found to be less than those obtained by experiment (19). Structural change may also arise due to damage by the electron beam or the requirement to operate in high vacuum. Noncontact AFM has the tremendous advantage of operation in air with no sample preparation. Generally, care has to be taken in the use of AFM for determination of pore sizes as the measured pore dimensions are a result of the convolution between tip and pore shape, which can become complex if the tip and surface pore sizes are comparable. However, this effect is expected to be much less significant for noncontact AFM than contact AFM as in the former case the tip is held at a significant distance from the surface and responding to a variation in attractive forces on scanning. (A detailed theoretical analysis is in progress.) Nevertheless, there is still a possibilty, especially for rough surfaces or membranes where the pores lie clustered in depressions, that for funnel shaped pores AFM is measuring a surface pore dimension at a lesser depth in the surface than the (underestimated) pore dimension determined by EM.

## 4. CONCLUSIONS

Noncontact atomic force microscopy is an effective means of examining the surface structure and pore morphology of porous structures such as ultrafiltration membranes. Samples can be examined by AFM without preparative procedures that might alter the membrane structure. The surface topography of membranes can then be explored without distortion when the membrane is in air. Image analysis can provide detailed information on the surface pore structure and allows quantitative determination of the pore size distribution. Hence, noncontact AFM can be used to determine surface pore sizes of membranes made from sensitive materials such as regenerated cellulose. Such materials may not be imaged directly by electron microscopy without damage caused by even very low energy electron beams. Pore size distributions obtained by noncontact AFM are comparable to those obtained by contact AFM, but differ significantly from those obtained by electron microscopy. Further work is needed to unambiguously resolve these differences. However, the technical benefits of AFM, combined with the ease of use of the technique, suggest that it should be considered as one of the methods of first choice for the determination of the surface pore characteristics of ultrafiltration membranes. In particular, imaging in liquids opens the exciting prospect of visualization of such membrane surfaces under their conditions of use.

## ACKNOWLEDGMENTS

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